12. Create and configure an application on Google App Engine and deploy a simple web application to understand its environment setup, deployment process, and basic working in cloud computing.
OR
Installation and Configuration of virtualization using KVM.

You'll usually get **one of these two options** in the exam. I'll prepare you for **both**, but focus more on **Google App Engine (GAE)** because it's more commonly asked and easier to demonstrate.

✅ Option 1: Google App Engine (Recommended)

🌐 What is Google App Engine?

- Google App Engine
- A **Platform as a Service (PaaS)** that lets you deploy web apps without managing servers

👉 Viva line:

> It allows developers to deploy applications without worrying about infrastructure.

🧪 Steps to Create & Deploy Web App

1️⃣ Go to Google Cloud

- Open: https://console.cloud.google.com
- Sign in with Google account

2️⃣ Create a Project

- Click **New Project**
- Give name → Create

3 Enable App Engine

- Search **App Engine**
- Click **Create Application**
- Select:
  - Region (e.g., Mumbai)

4 Install SDK (if using local system)

Install:

- Google Cloud SDK (`gcloud`)

👉 In lab, sometimes already installed

5 Create Simple App (Python Example)

Create file: `main.py`

```python
from flask import Flask
app = Flask(__name__)

@app.route('/')
def hello():
    return "Hello, App Engine!"

if __name__ == '__main__':
    app.run()
```

Create file: `app.yaml`

```yaml
runtime: python39
```

6 Deploy Application

Run:

```
gcloud app deploy
```

Then:

```
gcloud app browse
```

🎯 Output

- Web page opens:

```
Hello, App Engine!
```

🧠 Viva Questions

❓ What is App Engine?

> PaaS service to deploy apps without managing servers.

❓ What is `app.yaml` ?

> Configuration file defining runtime and environment.

❓ Advantage?

> Auto scaling, no server management.

⚠️ Common Mistakes

- Not enabling App Engine first
- Wrong runtime in `app.yaml`

- Forgetting to deploy (`gcloud app deploy`)

🔁 Option 2: KVM Virtualization (Alternative)

💻 What is KVM?

- Kernel-based Virtual Machine
- Turns Linux into a **hypervisor** (like VirtualBox)

👉 Viva:

> KVM allows multiple virtual machines on a Linux system.

🧪 Installation Steps (Ubuntu)

1 Install KVM

```
sudo apt update
sudo apt install qemu-kvm libvirt-daemon-system virtinst bridge-utils
```

2 Check Installation

```
kvm-ok
```

3 Start Service

```
sudo systemctl start libvirtd
```

4 Create VM

- Use:
  - `virt-manager` (GUI)

    OR
  - command line

5 Install OS in VM

- Load ISO file (Ubuntu/Windows)
- Allocate RAM, CPU

🧠 Viva Questions

❓ What is KVM?

> A Linux-based hypervisor for virtualization.

❓ Type of virtualization?

> Full virtualization

❓ Advantage?

> Efficient, open-source, uses hardware virtualization

🎯 Which One Should You Choose?

Option	Difficulty	Exam Preference
App Engine	⭐ Easy	⭐⭐⭐ Highly preferred
KVM	⭐⭐ Medium	⭐ Sometimes asked

👉 Go with App Engine if choice is given